

Team J8 Autonomous Retrieval Design

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Cora Ilic, Tina Song, Jeremy Yu, and Layan Younis**

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**The Challenge:
Design a fully
autonomous
retrieval system to
pick up designated
objects**



Our Team's Strategy:

Process:

- Each team member created prototypes individually
- We initially wanted to make multiple claws (for risk management) but decided against it due to material constraints

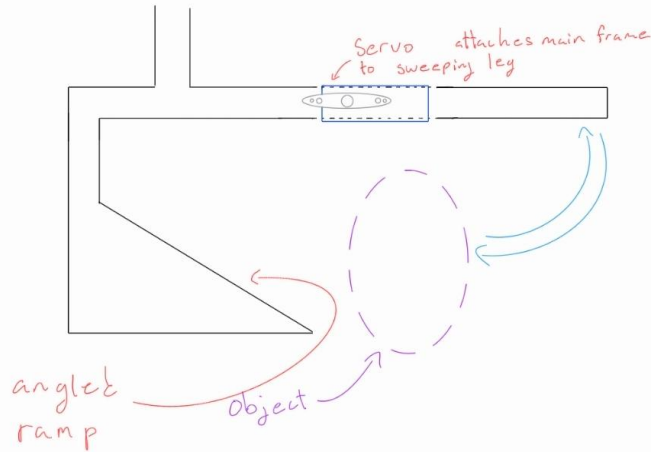
Competition Day:

- Focus on speed, communication and trust amongst team members

Concept Generation & Initial Prototyping:

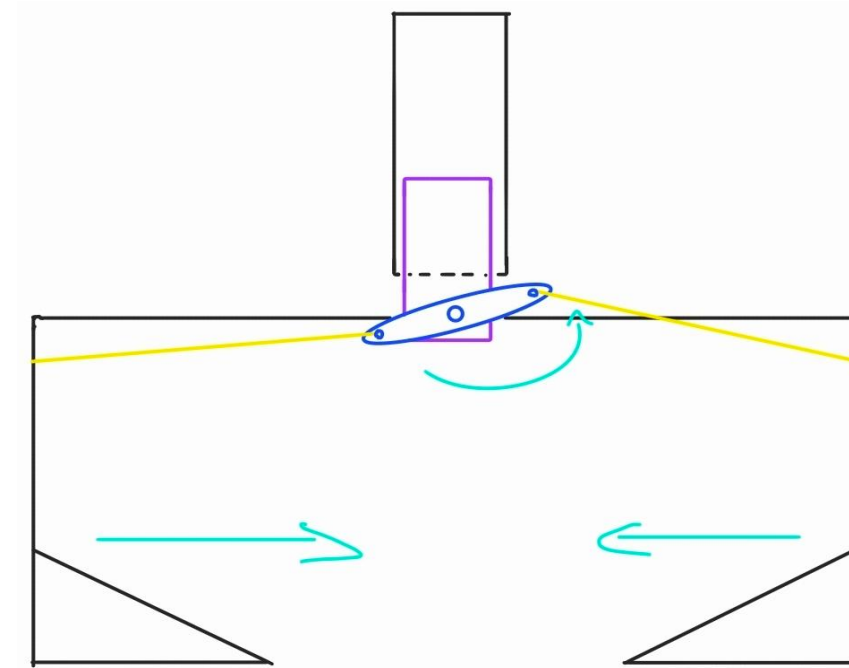
The Sweeper (JG)

- Broom like design
- Pushes items in with rubber band sprung door



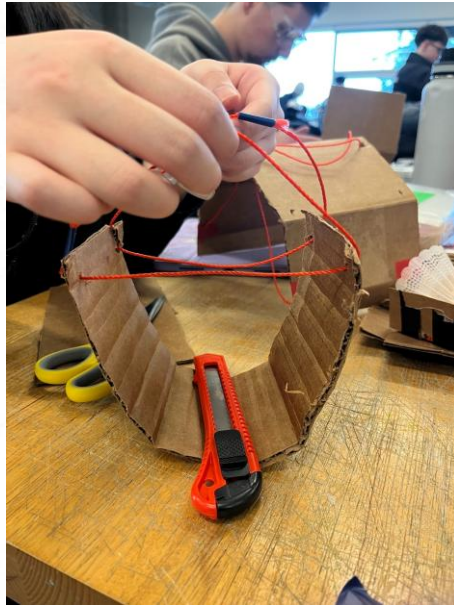
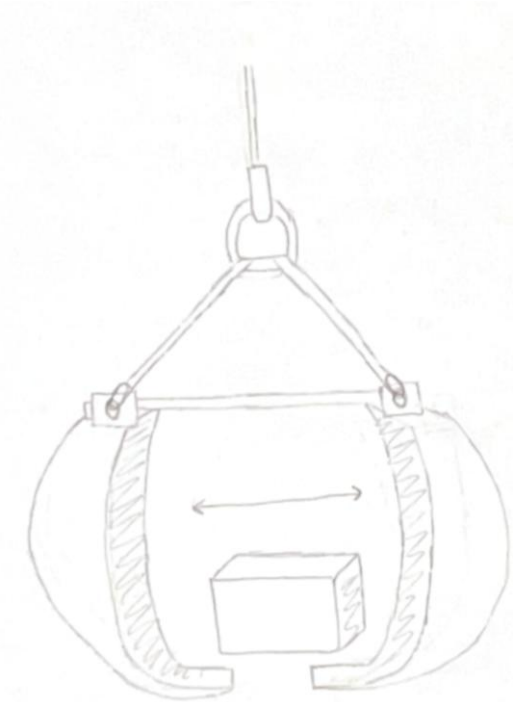
Trapper (JG)

- Creates a closed off space to store the objects
- Pushes items onto each ramp eventually closing



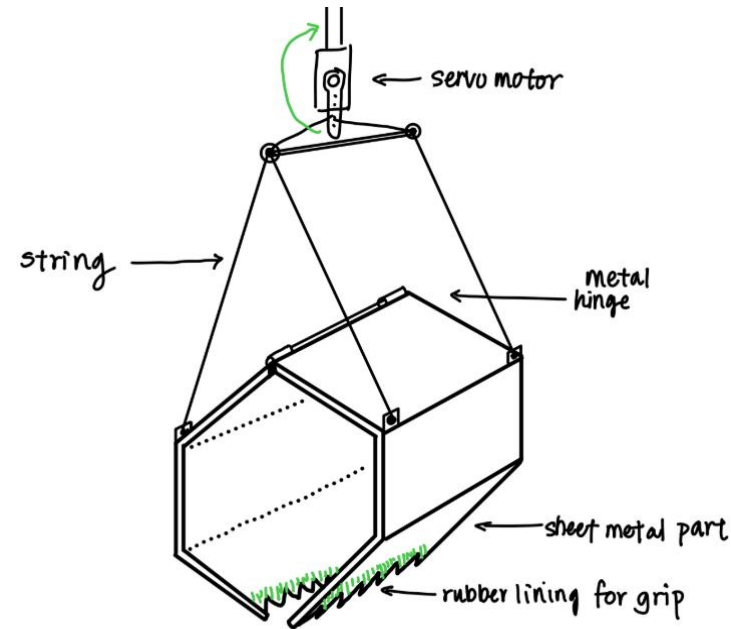
Clamshell Grip (LY)

- Curved arms
- Strings open and close the claw
- Would be more stable if the claw is connected at the top



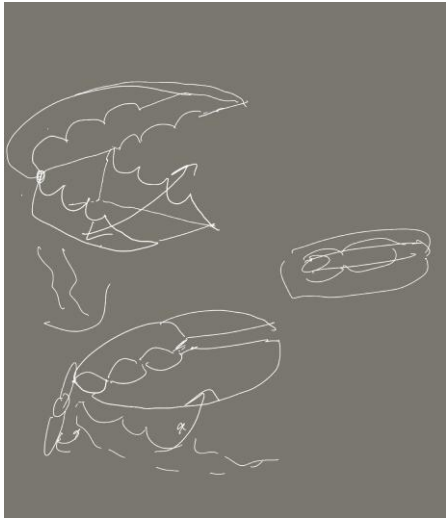
Hexagon Claw (TS)

- Two pieces connected with a hinge
- Strings on either side bring the teeth together when pulled on
- Strings placed closest to the opening of the claw make it easier to operate



Elastic Grip (JY)

- Rubber bands lock around objects
- Servo would need more force to counter bands
- Sparked assisted opening rather than closing used in final design



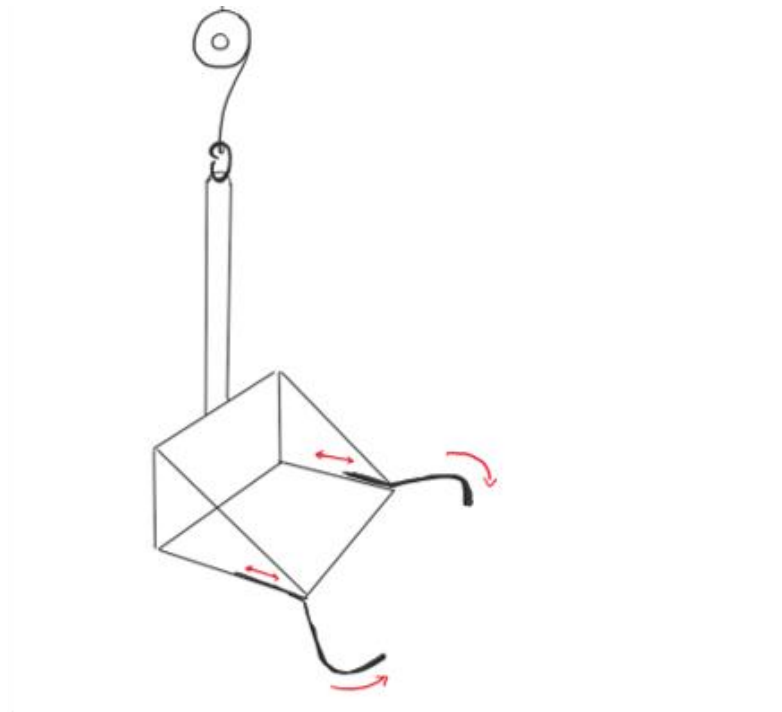
Pulley Claw (JY)

- Not optimal for transferring torque
- Sparked string and servo connection used in final design



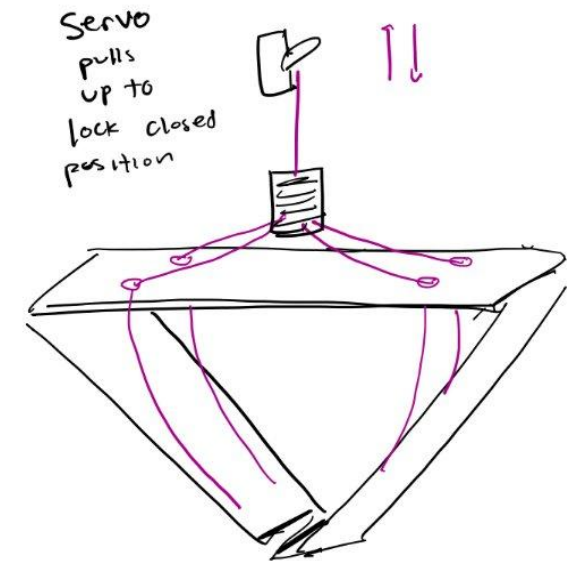
Claw Pan (CI)

- Dustpan shaped with hooks to secure item onto the pan
- Difficult to operate both claws with one servo



Gravity Strings Claw (CI)

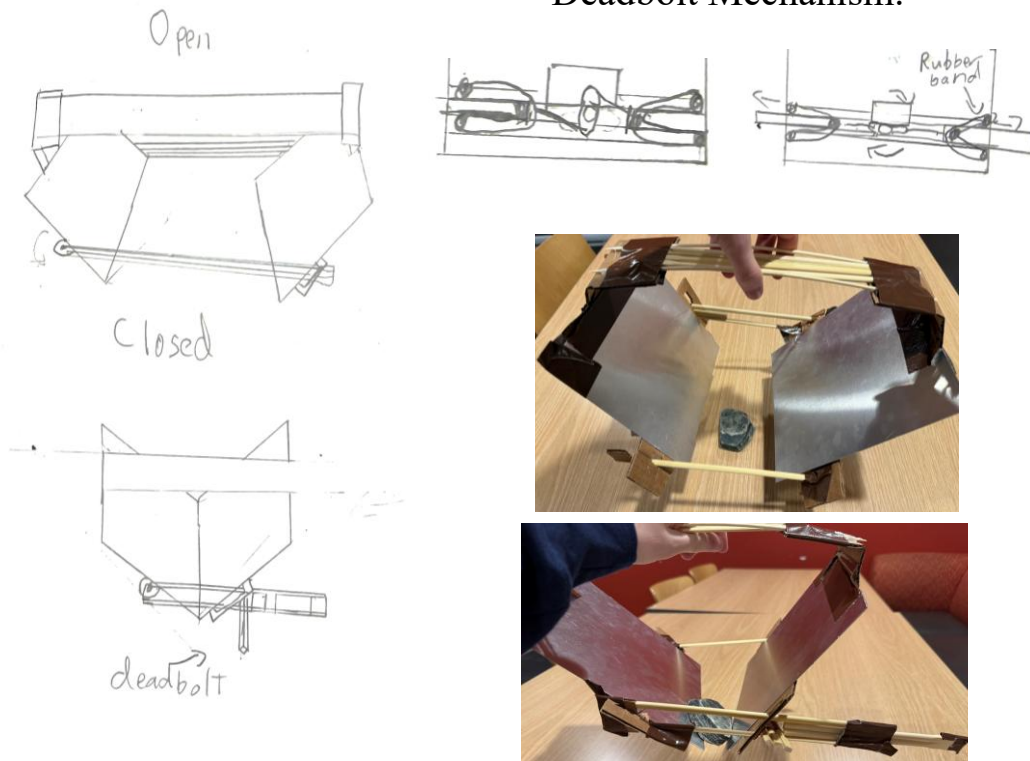
- Naturally closes due to weight and hinge angle when lowered to the floor
- String mechanism limits weight capacity of servo



Gravity Deadbolt Claw (EF)

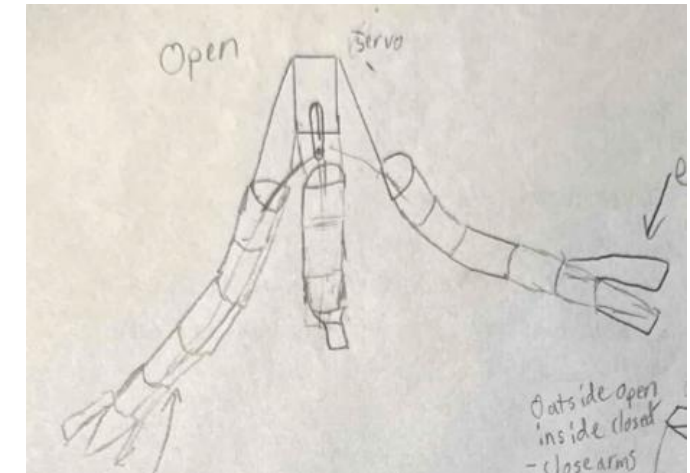
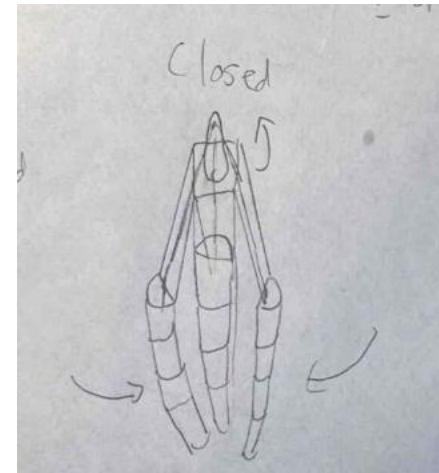
- Shallow claw to floor angle for easy scooping.
- Uses momentum of dropping and weight of claw to force itself closed upon touching the ground
- Deadbolt mechanism minimizes torque on servo.
- Deadbolt too inconsistent

Deadbolt Mechanism:



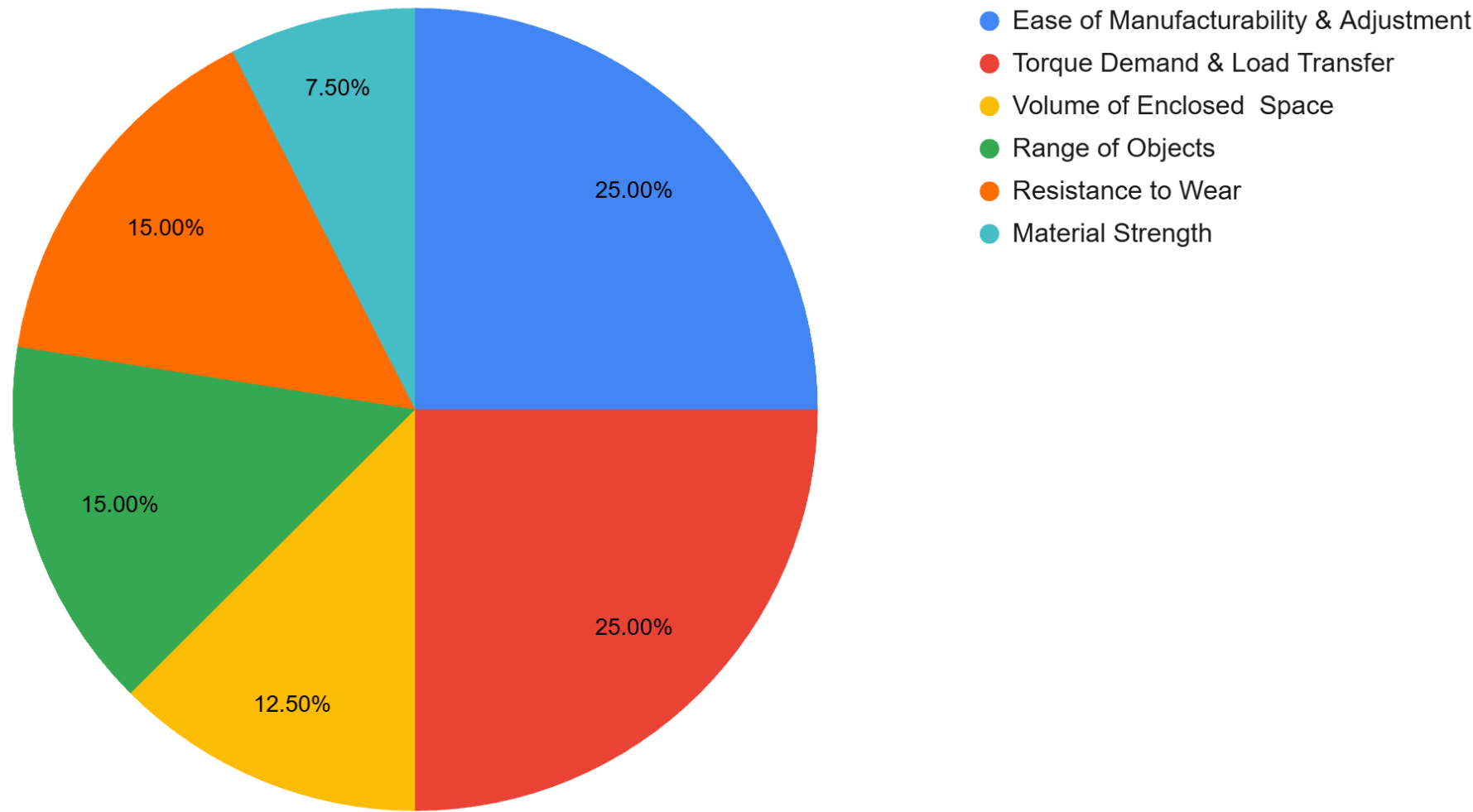
Octopus Slider (EF)

- Servo pulls tension in limbs to close arms.
- High degrees of freedom, frequent problems
- Many links increases build time
- Can't pick up small items.



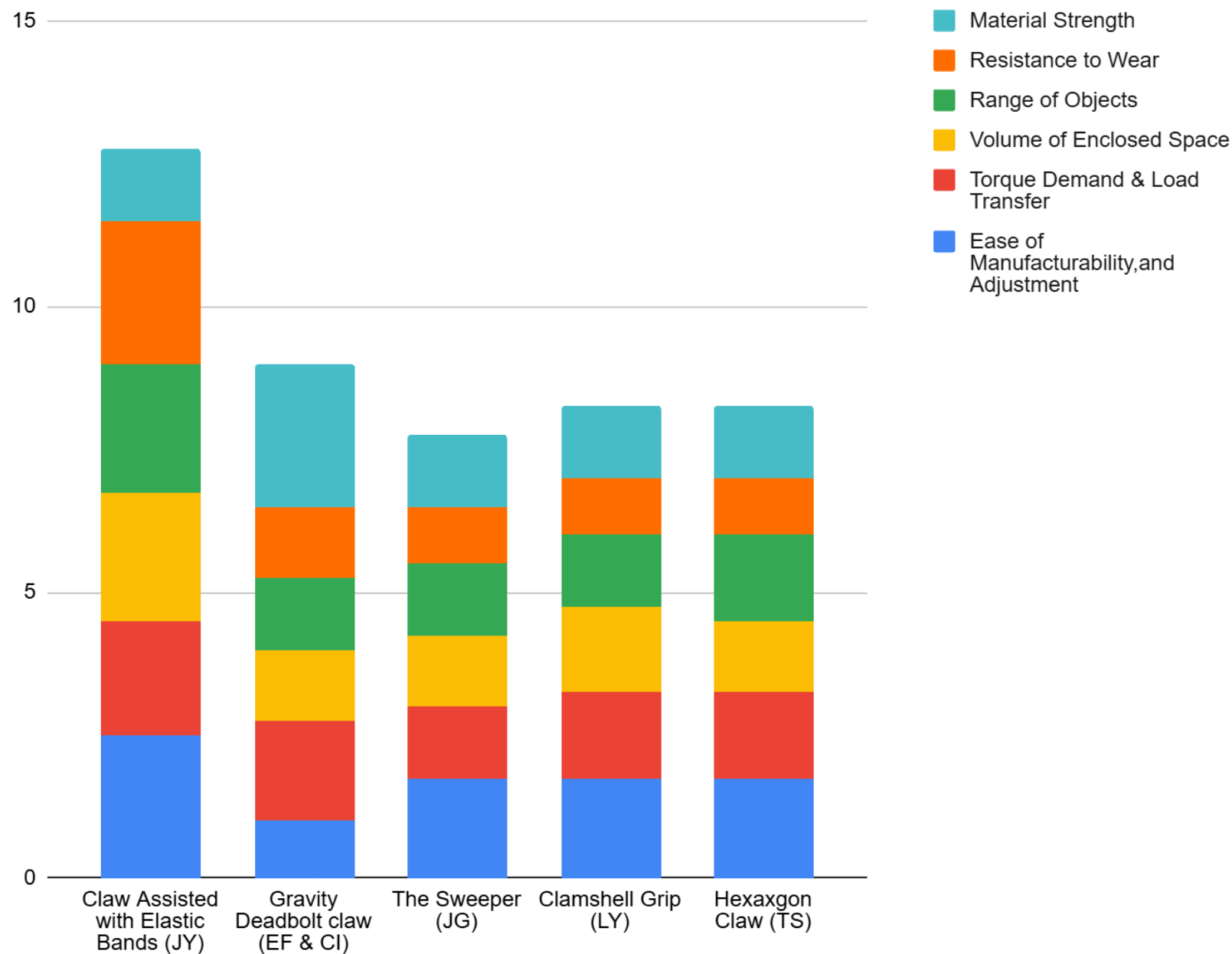
Scoring – WDM Weight Distributions

Pie Chart of WDM Weight Distributions



Scoring – WDM Results Per Design

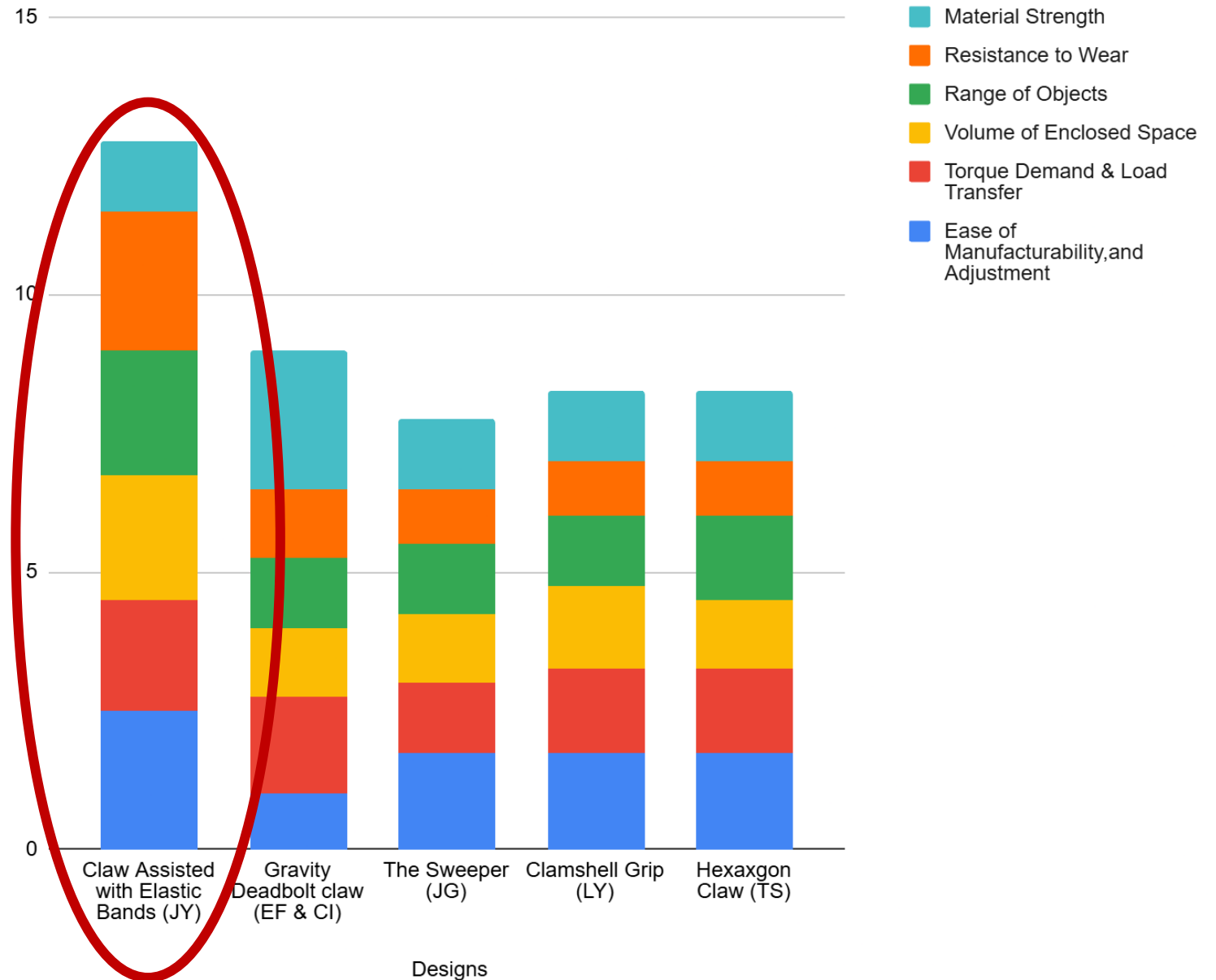
Stacked Bar Chart for WDM Results for Proposed Designs



Scoring – WDM Results Per Design

No sensitivity analysis
needed – clear best design.

Stacked Bar Chart for WDM Results for Proposed Designs



Risk Identification + Analysis

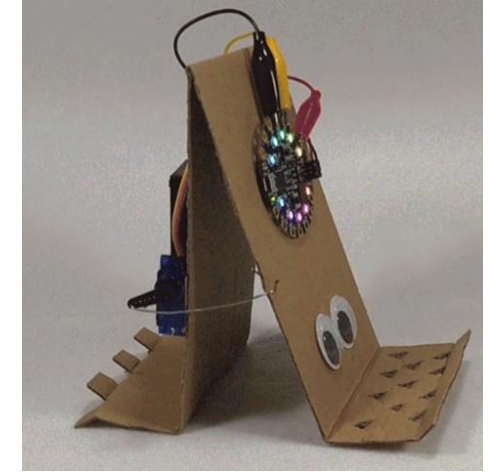
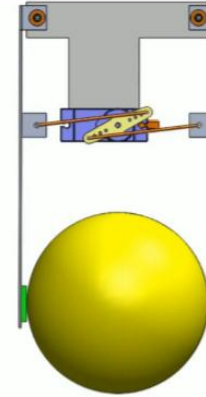
We identified strategic, operational, and technical risks. The two most significant risks were analyzed:

1. Sketches with servo placement need to be validated

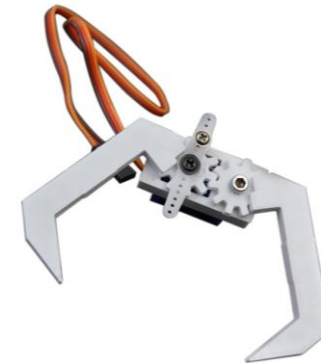
- We don't know if the servo placed on a sketch will work under real-world conditions

2. Fixating on mechanical transmission systems that allow for stronger grip

- Time wasted and no claw made if final design has too many frictional losses



(APSC 101 Module 5 Skeleton Notes)
Different possible placement for servos to close things



(Walmart)
Claw above uses a gear system. But we don't know if it works well if made with allowed materials.

Investigation 1: Servo placement

Investigation Method:

- Test alternative servo placements
- Calculate and compare online values of torque rating
- Build a physical prototype to test for
 - Deformation of material
 - Stalling of servo

Successful test:

- Virtual: torque of our design $<$ maximum torque
- Physical: Elastic deformation - functionality is maintained

Investigation 2: Transmission systems

Investigation method:

- Try building gears by layering cardboard
- Build a simple belt-and-pulley to test different pulley size ratios and belt/axle materials
- Observe which setups minimize friction and reduce required torque

Successful Test:

- Minimal slipping/friction



(Storm The Castle)
Example pulley
system using
cardboard and string

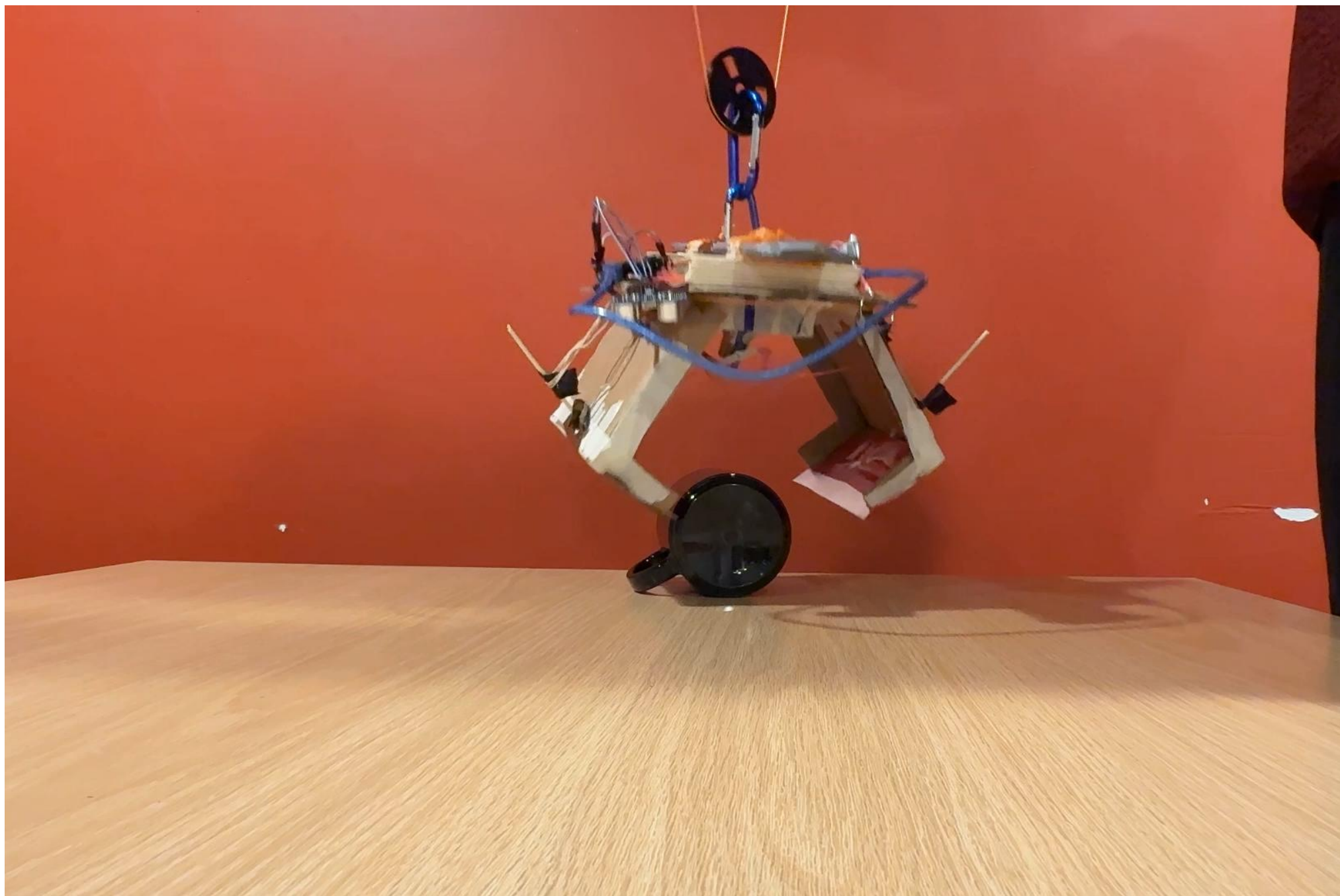
How these risks affected our design:

- Servo was found to be sufficiently strong to close an open claw even if it was resisted by rubber bands
- Gears could not be made with the permitted materials
- Pulleys were too unstable and the belt would break easily
- Transmission systems were abandoned in our final design

Final Design

- Combined the bar from the Gravity Deadbolt Claw with a new box-like claw
- Operated with a servo motor which pulls on strings attached to the inside of the claw
- Rubber bands help pry claw open when strings are slack

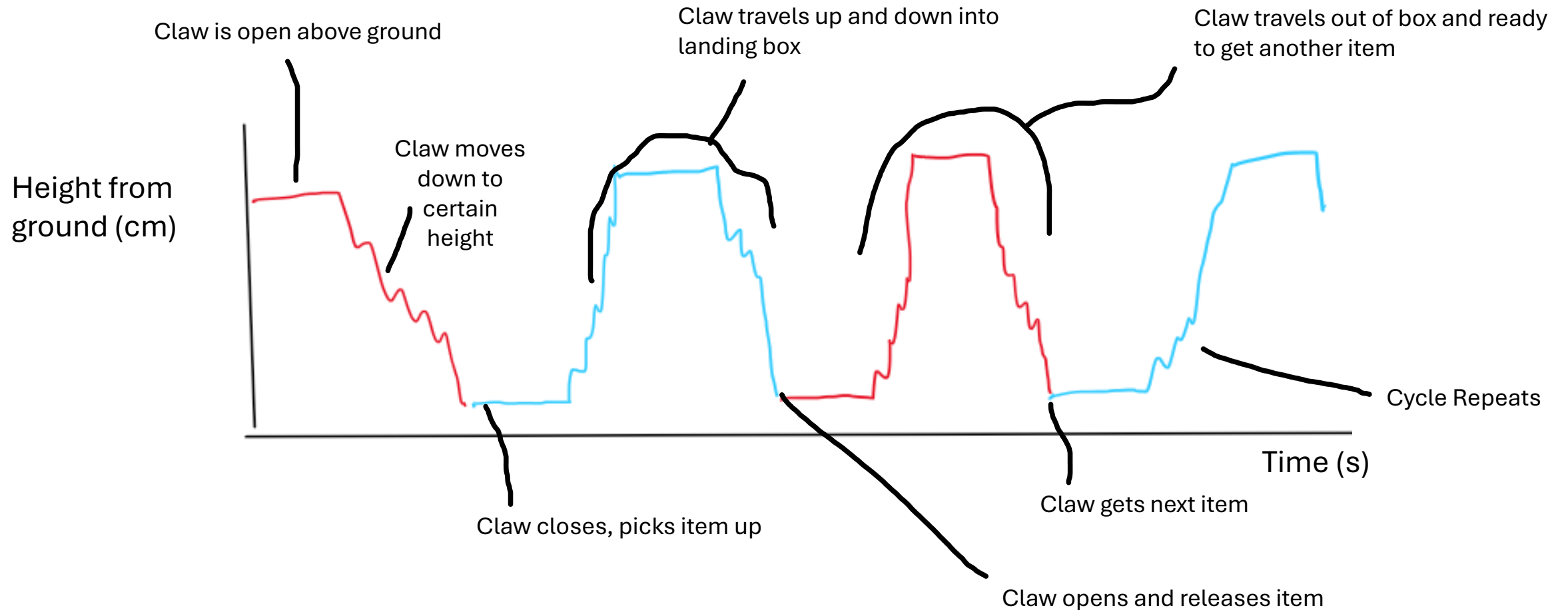




Final Code Implementation – two states

- Our team did not want to rely on timer delays, so we implemented two states – OPEN and CLOSED
- Needed to find a different way to toggle between states
- Found by observing Height of Claw vs Time
 - Height was given by data from ultrasonic sensor
 - Found critical regions to indicate when to change between the two states

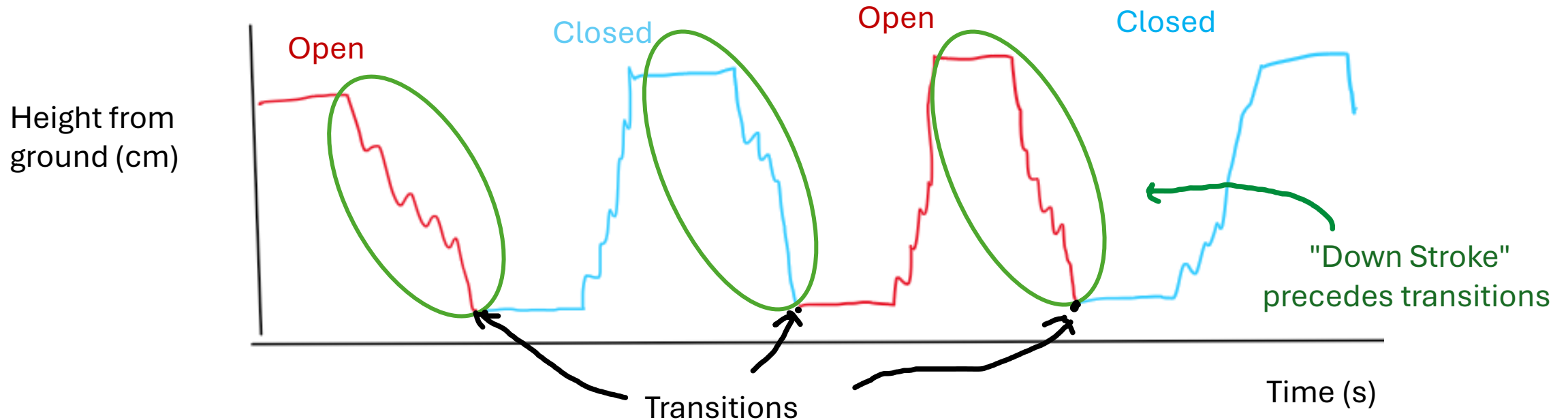
Final Code – Graph of height dependent claw actions during object retrieval



Final Code – Height vs Time, Finding Transitions

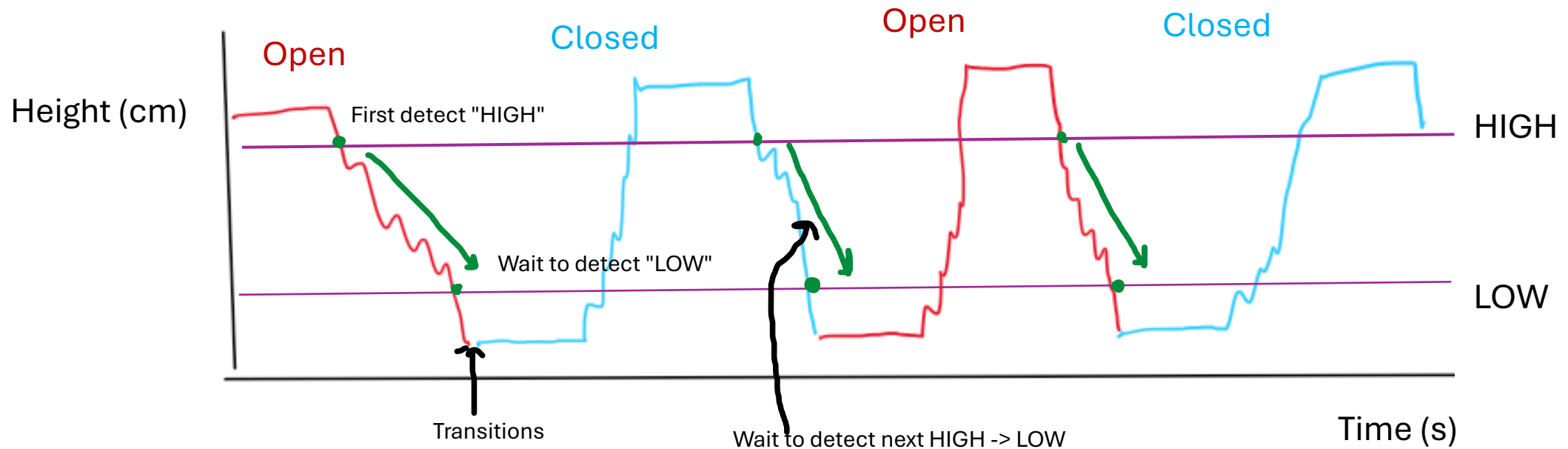
Removed labelled actions to just states, and found transitions.

Insight: State needs to transition **after** a "Down Stroke" is detected!



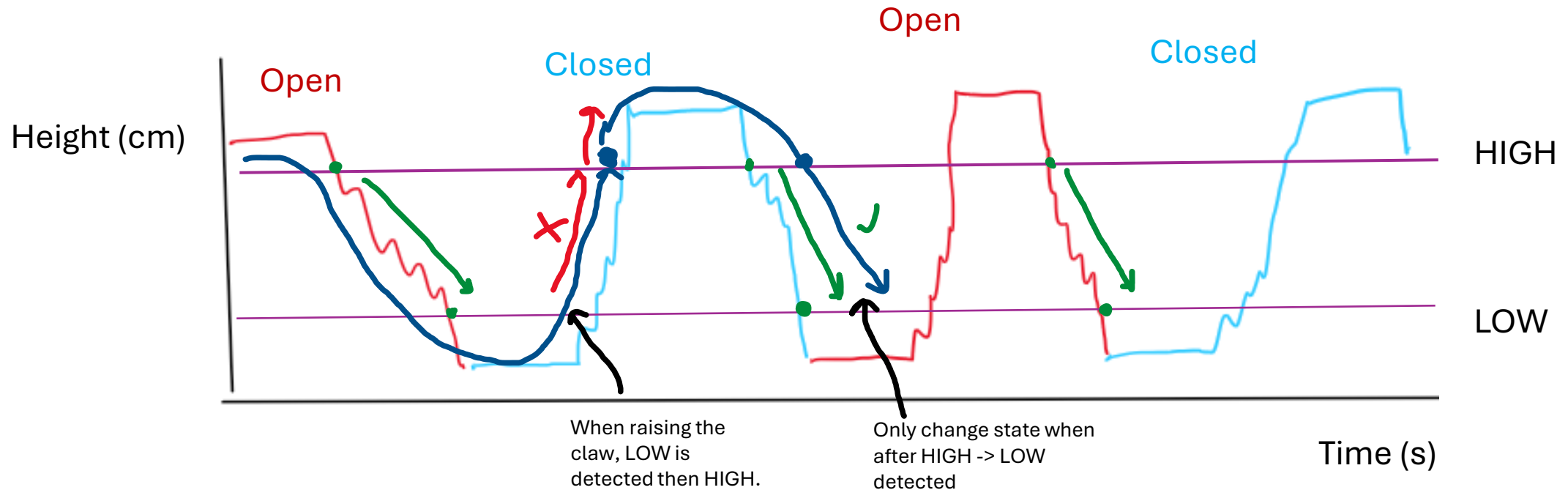
Final Code – Implementation

"Down Strokes" can be implemented as ordered height detection between two arbitrary thresholds!



Final Code – Robustness

Though, HIGH is detected when going up, the claw state doesn't change **since LOW was detected first**.

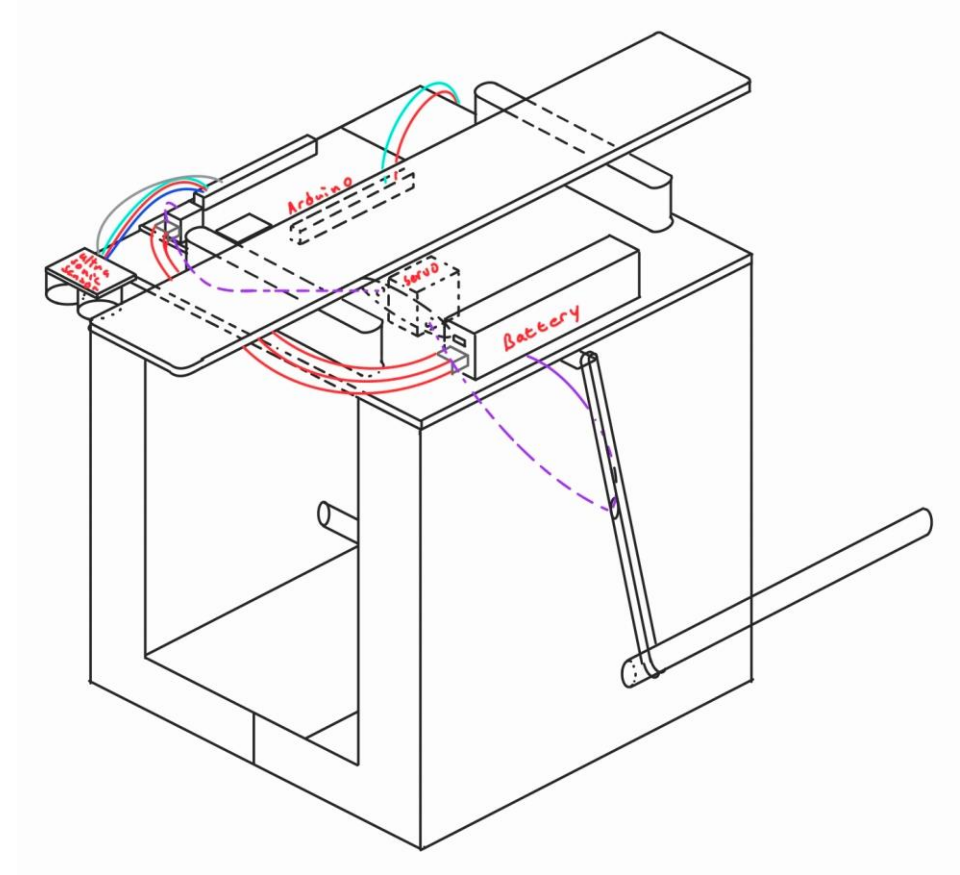
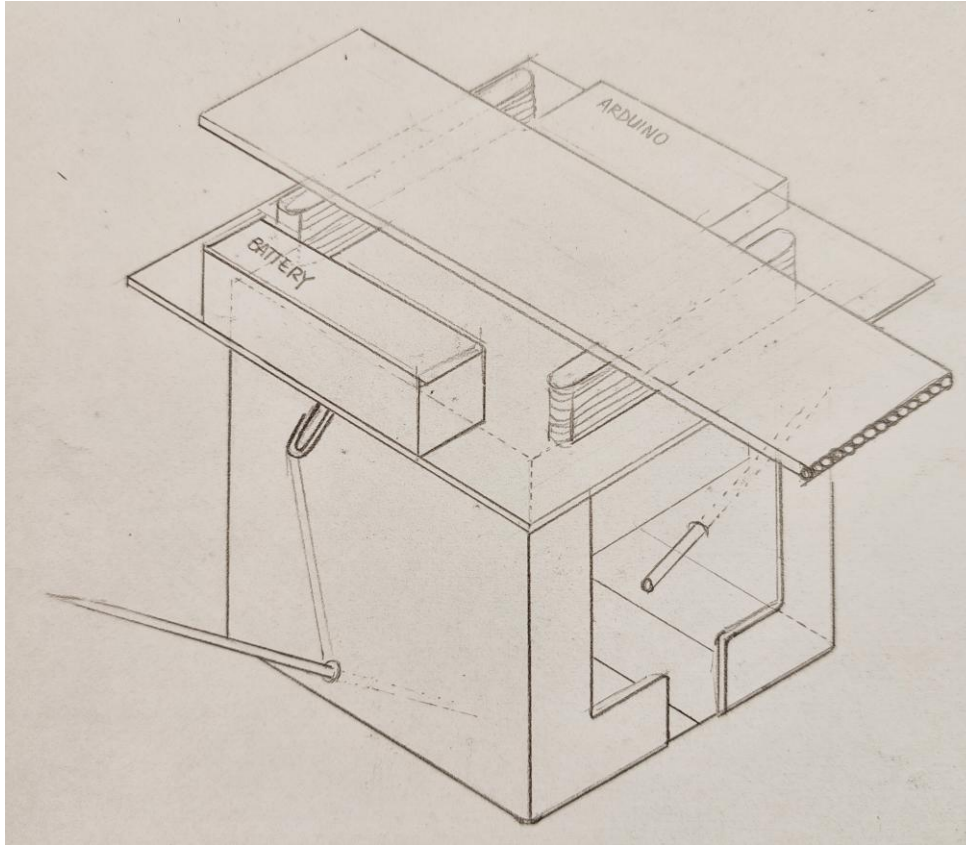


So, the claw never changes state when going up. **HIGH must be detected before LOW** to trigger a state change.

Final Code – Modifications from Experimentation

- Smoothing of data was added from ultrasonic sensor
 - This disregarded noise from doppler effects (moving too fast!)
- Thresholds, HIGH = 40cm and LOW = 16cm were set
 - Accounted for the dimensions of the claw and placement of sensor relative to ground

Final Design Isometric Drawings



Competition Results + Reflection:

Competition results were consistent with our risk analyses, prototyping and experiments:

- Round 1 Results: Many different objects were picked up without issues
- Round 2 Results: 21 blocks were gathered
- Our claw performed better than many timed delay claws, which confirmed our choice of height detection based code
- No groups used pulleys, which supported our analysis that they were unstable and subject to more friction

Contributions:

Slide 4 – Team Strategy : Cora Ilic

Slide 5-9 – Sketches and Prototypes: All group members

Slide 10-12 – Selection Process : Jeremy Yu

Slide 13-16 – Risk Analysis/Investigation : Tina Song, Layan Younis

Slide 17 – Final Design: All group members

Slide 18 – Final Claw: Evan Fielding

Slide 19-24 – Final Code: Jeremy Yu

Slide 25 – Isometric Drawings - Jacob Gupcsi, Tina Song

Slide 26 – Competition Results/Reflection – Jeremy Yu, Tina Song

**No Generative AI was used in the
production of this presentation.**